

Ohm VR: solving electronics escape room challenges on the roadmap towards gamified STEAM education

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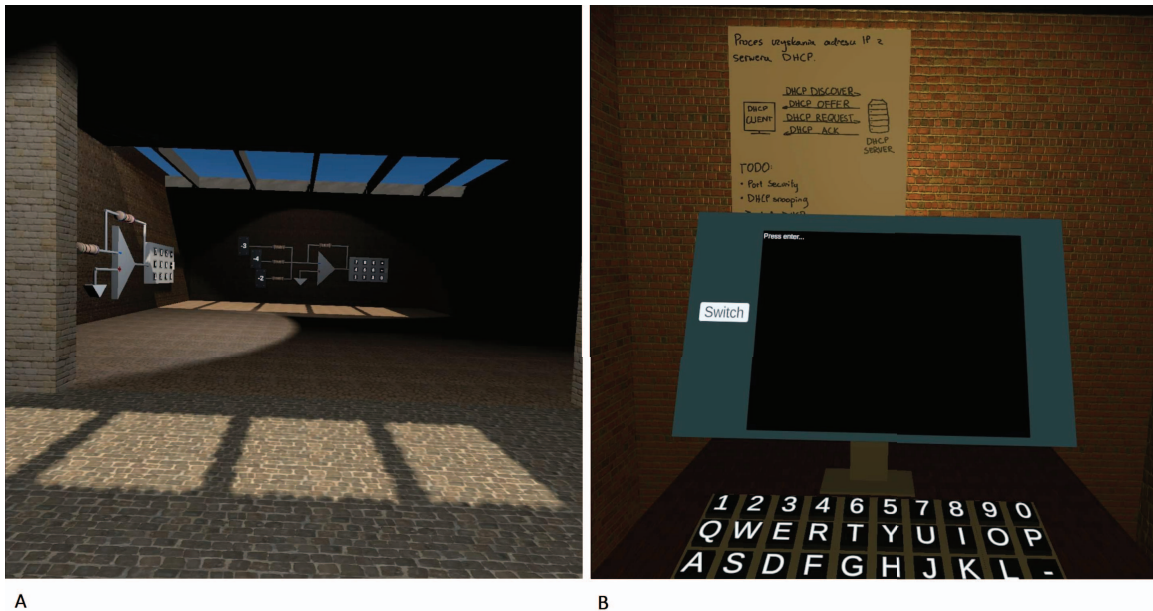


Figure 1: Locations in the escape room game: A - Level 2, B - Level 3.

ABSTRACT

We present a prototype of VR application containing puzzles inspired by Electronics and Telecommunication classes validated qualitatively. The design utilizes escape room mechanics and the application features 3 levels: solving resistors circuits, amplifier circuit and the network configuration. It is implemented with Unity for Meta Quest 2 platform. The experience lasts about 30-45 minutes. It was evaluated by usability and user experience tests conducted with N=5 users. Learning outcome and knowledge retention was measured with a pre-test and post-test.

Index Terms: Virtual reality—Education—Gamification—Electronics; STEAM—Escape room—Visualization design and evaluation methods

1 INTRODUCTION

The motivation for this project was to explore the potential of using gamified Virtual Reality experience for teaching selected basics of electronics and telecommunication. This project is part of the

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experiential learning trend, modeled, among others, in the form of the Kolb cycle [23] and the Edgar Dale cone of experience [12].

COVID-19 pandemic induced the necessity of remote education, which opened some new opportunities and acceptance for using simulators instead of in-person laboratory exercises. According to the insights collected from students, one of the barriers in studying at the university is an anxiety of being constantly assessed for the mistakes and errors. Another problem connected with the methods used so far in education is the insufficient involvement of students in active experience, experimentation and reflection resulting from observing the effects of their actions, which, combined with reduced states of concentration in the era of social media, results in low knowledge retention. As it was stated in [11], educators must quickly navigate and leverage the differences between traditional simulation and telesimulation to create robust remote educational experiences.

The trend is visible almost at every University. At our University there were several approaches to design various VR tools to support both knowledge acquiring [20] as well as learning soft skills [15], [14].

VR experiences that make the user feel safe and are characterized by high immersion and interactivity have the potential to fill the existing gap. Such tools, however, should first be subjected to detailed validation, to which this project is intended to contribute. It is worth noting that there is no consistency between research published in recent years on the impact of immersive and interactive environments on learning, with both positive [16], using functional

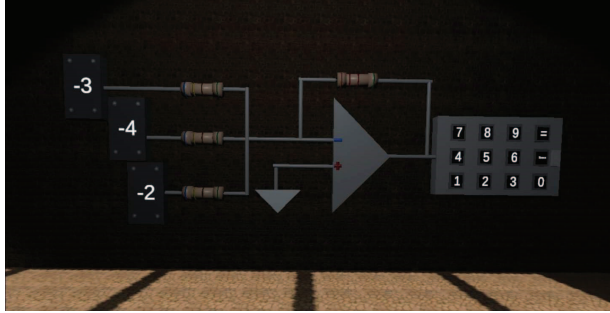


Figure 2: A puzzle in the escape room game - a circuit from level 2.

magnetic resonance imaging (fMRI) experiment and, surprisingly, even negative results [6].

Since the traditional escape room games have become widely known form of entertainment, we chose to apply this concept to engage users into immersive puzzle solving based on the domain which they are usually not familiar with.

2 STATE OF THE ART

The opportunities brought by VR to the engineering education have been pointed since over a decade [5], also in the field of Electronics [9, 10]. According to the recent research [22], students using VR had better learning results compared to traditional methods. It is suggested that there will be no need to maintain big and expensive laboratories if something can be done using just a headset. Teaching engineering knowledge has been a subject for various attempts of gamification. Projects e.g. by *Labster* [4, 26] create virtual laboratories twinning the traditional ones. Games like *while True: learn()* [18] which introduces machine learning issues or *Hacknet* [25] have met extremely positive feedback from community, while others, like *Electronics Circuits Simulator* [8] obtained mixed opinions. Some of the recent productions are multiplatform, like *CRUMB Circuit Simulator* [2].

For VR platform, *Short Circuit VR* [3] is one of the electronics lab simulators presented in Virtual Reality. *Short Circuit VR* [3] makes perfect use of the full dive, providing both immersion and UX reflected by overwhelmingly positive Steam opinions. Another example is *Warz - Voice Driven Virtual Assistant Tutor and Simulator in Virtual Reality* [1] for teaching fundamental electronic and electrical circuit behavior and operation. Recent *ELIAKI VR* project [19] describes design of entire Industry 4.0 laboratory in the multiplayer mode.

Our contribution to this trend is a first proof of concept in exploring the widely known escape room mechanics to make the electronics puzzle a gamified part of the educational VR experience.

3 DESIGN

This project has been built as an application for the Meta Quest 2 headset. Using hand tracking highly improved the immersion of certain interactions, eg. typing on the keyboard. There are levels where controllers are used for movement, and for object manipulation including button pressing, but there is also a whole level where player does not need controllers. To prevent VR motion sickness, project uses teleportation mechanics to let player move around the virtual environment. This project was developed using Unity environment. Applied graphic effects and textures are from [13].

3.1 Level 1

Level 1 is designed to familiarize the players with the game so that they can adapt and learn the provided affordances. Spawn point of the player is located in closed room with dimmed light. On the walls

there is electronic circuit with five slots. Player has to match resistors THT in correct order to get required result. Notes are available to help player with understanding of the electronic circuit. Notes introduce information about series equivalent resistance, parallel equivalent resistance, the appropriate colors of the stripes on the resistances and voltage divider.

3.2 Level 2

Level 2 is created to introduce relatively simple concepts of analog circuits electronics. The participant of the game is not obliged have advanced knowledge in the field of construction of operational amplifiers from basics i.e. how it is built of transistors and other elements electronic. The background extension may also include issues from the CMOS or bipolar transistors. In this way, the knowledge of the participant can be increased operation of transistors.

3.3 Level 3

Level 3 is prepared to make player more familiar with computer networks. The problem to solve is preparing correct configuration of switch which will contain DHCP snooping and DHCP configuration. Player is alone in small room with just a desk with PC on it. They are supposed to use their own hands in this level because there is a hand tracking functionality which allows to type on keyboard with better performance. Network should be protected against DHCP Discover attack which can lead to exhaust the pool of IP [21]. That kind of attack is really dangerous for network because in case of empty pool new devices are not able to connect network and that makes network totally useless. The main goal of network is obviously a possibility to connect with others. To complete level there should be a correct rate limit set on switch, enabled DHCP snooping and DHCP server should be trusted on switch to protect from fake DHCP injection to network.

4 EVALUATION

4.1 Methodology

The prototype was initially evaluated with scenario-based usability tests. A triangulation of qualitative and quantitative was applied. N=5 (2 female, 3 male, mean age 28.2, standard deviation 12) users were investigated, among them only there was only one student of Electronics and Telecommunications, two of them had engineering background from other disciplines and the rest two had no education in Electronics at all.

All of them completed the scenario, the time of completion varied between 30 and 45 minutes.

The procedure consisted of:

1. Before VR experience:

- participants signed an informed consent form;
- introductory test was performed to assess demographic profile of the research group as well as the previous experience with games and VR;
- SSQ questionnaire [17] in order to assess susceptibility to the simulator sickness ;
- pre-test 1 to assess the initial state of knowledge on the issues used in the game. It consisted of 10 questions, 6 of which were single choice, and 4 of them were multiple choice. The maximum score was 13 points.

2. During VR experience:

- VR experience and usage of controllers was introduced to each participant;
- the participant went through the experience, using the thinking aloud protocol.

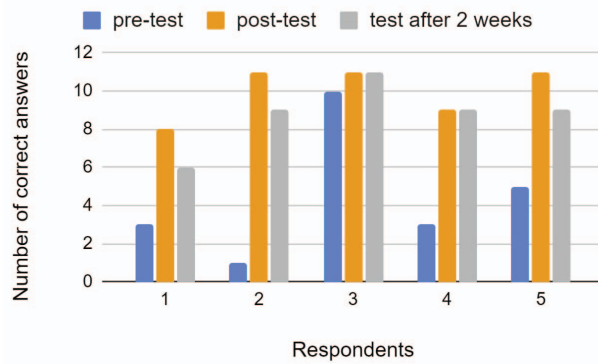


Figure 3: Learning outcome and knowledge retention for 5 users: blue - score in the pre-test, orange - score in the post-test, grey - score in the test after 2 weeks.

3. After VR experience:

- Post-test of the knowledge was performed in the same form as the pre-test;
- SSQ questionnaire was used to make sure if any symptoms of simulator sickness were experienced;
- System Usability Scale (SUS) questionnaire was performed in order to assess the usability of the solution as a complete system. The tool designed in [7] was selected because of its universal nature in the investigation of various digital systems ;
- Polish adaptation of the Immersion questionnaire was used to measure the experience of each participant [24];
- Semi-structured discussion was conducted at the end of the session in order to collect the deepened impressions and reflections of the participants on the game they experienced as well as possible further directions of development.

The same test was performed once more after 2 weeks from the day of participating in the Ohm VR experience.

4.2 Results

Although the numerical results cannot be interpreted quantitatively with statistical significance because of the small population, we will present the results to support the qualitative interpretation. Knowledge of the educational topics used in the VR game has increased for every participant (Fig. 3), before the experience mean score was 4.4 but it varied within the group a lot (standard deviation 3.4), and after the experience - the average was 10 (standard deviation 1.4). In the open discussion majority of participants declared that the game was difficult to them, yet all of them completed it and expressed the satisfaction of solving the puzzles. After 2 weeks, mean score was still high (average 8.8 with standard deviation 1.8).

Mean SUS score was 62.5 points, which can be interpreted as a satisfactory as for the first iteration of the prototyping. Immersion questionnaire showed that the participants put a lot of effort in the game and were curious of the further challenges. They declared that they were highly focused and immersed.

5 DISCUSSION AND FUTURE WORKS

The mechanics of escape room was enthusiastically perceived by the testers as emotional and challenging, even for those who did not have an electronic or engineering background, which encourages for further exploring this concept and supplementing it with

Table 1: Test of knowledge. Questions with * are supplemented with an illustration not attached in this article. Questions 6-10 are multiple choice questions.

1	What is the value of the resistor? 52M Ohm, 520k Ohm, I do not know, 5.4k Ohm, 250k Ohm
2*	What is the value of a serial connection? 140 Ohm, 270 Ohm, I do not know, 680 Ohm, 410 Ohm
3*	What is the value of parallel connection? 4 Ohm, 300 Ohm, 80 Ohm, 500 Ohm, I do not know
4*	What voltage will be at the output for an inverting amplifier? The input voltage is -1V. What will be the output voltage for the amplifier which is ideal? I do not know, 100, 600, 6, -6
5*	What voltage will be at the output for a summing amplifier The output voltage is -3V, -2V and -4V respectively from top to bottom. What will be the output voltage for a summing amplifier configuration if the amplifier is ideal? I do not know, 6, 9, 54
6	How does a DHCP server work? - Provides MAC addresses to computers using DHCP, - It provides, among others, IP addresses to computers using the DHCP protocol, - Allows you to change the website address to an IP address, - It allows you to send frames between devices in a given computer network, - I do not know.
7	Choose all ways to secure the DHCP server - Cover with a warm blanket so that they do not lose heat - DHCP snooping, - Port Security, - Designation of a trusted DHCP server, - I do not know.
8	How does the switch work? - It is responsible for communication with external networks, - It performs the delivery of IP addresses through the DHCP protocol. - It is responsible for forwarding frames received on interfaces, - It uses a table of MAC addresses, - Creates a local network, - I do not know.
9	What are the steps to protect a DHCP server from a DHCP Discover attack? - Setting a password on user mode on the switch, - Setting the rate limit for DHCP Discover queries, - Setting the password to privileged mode on the switch, - Setting the mac address on the interface to sticky mode, - I do not know.
10	What does setting the mac address mode in the interface to sticky mean? - Allows manual entry of MAC addresses into Port Security configuration, - Allows you to specify the combined MAC address of several devices in the configuration, - Allows the MAC address to be automatically added to the Port Security configuration, - I do not know.

additional gamification methods, like time factor, performance point, leaderboards or skills tree.

The next step in the validation of this kind of experience would be the quantitative investigation with a larger group of participants in order to obtain statistically significant results. We plan to broaden the evaluation procedure to incorporate cognitive load measures.

Another path for further development is a multiplayer mode allowing for group learning or being accompanied by a teacher.

6 CONCLUSION

The prototype of VR escape room with riddles based on electronics and telecommunications knowledge was validated as a proof of concept. The three levels experience introduced and verified the knowledge on circuits based on resistors and amplifiers and the network configuration. It needs further development as well as a larger scale quantitative study to be improved in further works. The concept is modular and therefore it is easily scalable - it can be supplemented with further levels/rooms introducing further foundations of knowledge in the form of puzzles to solve. The crucial challenge in the designing this type of experience is the balancing and personalization the difficulty level to optimize the user flow.

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